EE-222 Data Structures and Algorithms

BS(CE)-2k21

LAB

REPORT#4&5

<u>**Submitted By:**</u>

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	**5**	**5**	**5**	**15**
Obtained				

Lab Instructor.
Kaynat Rana

Experiment #4 AND 5

Implementation of singly linked list (sorted and circular)

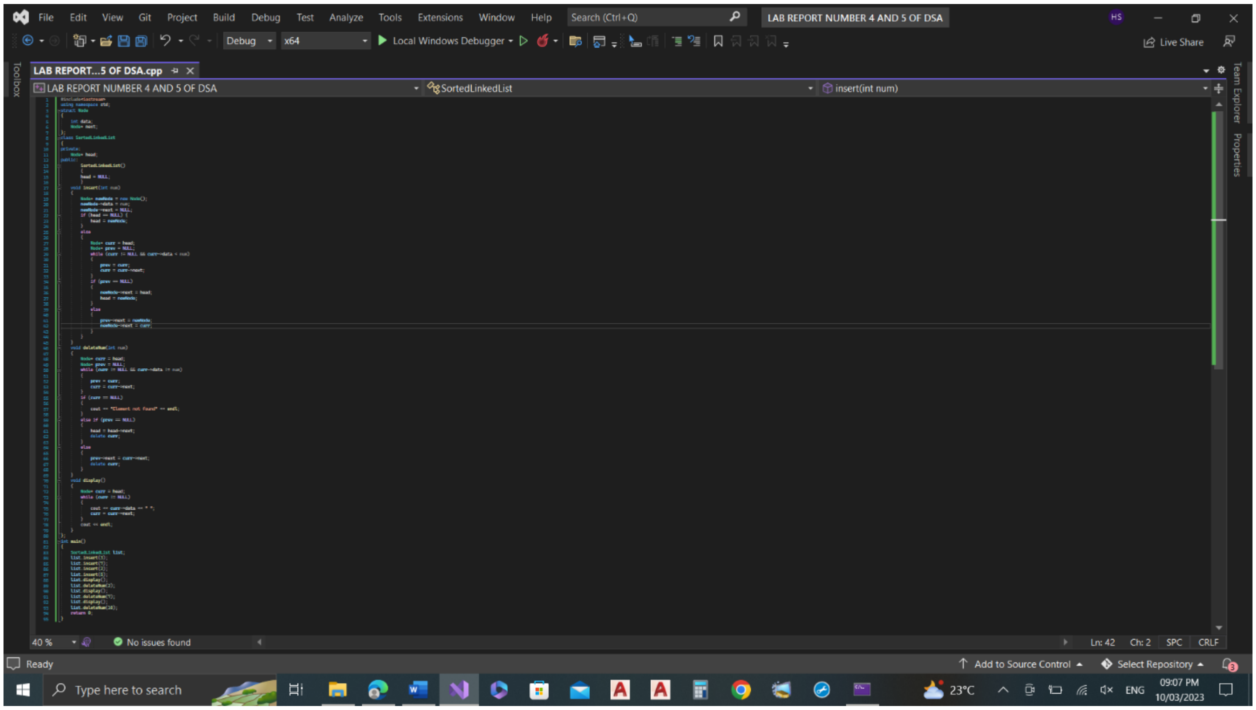
Objective:
The objective of this session is to understand the various operations on Singly linked list in C++.
1.Insertion
2. Deletion
3. Traversal

Equipment Used:
Visual studio C++

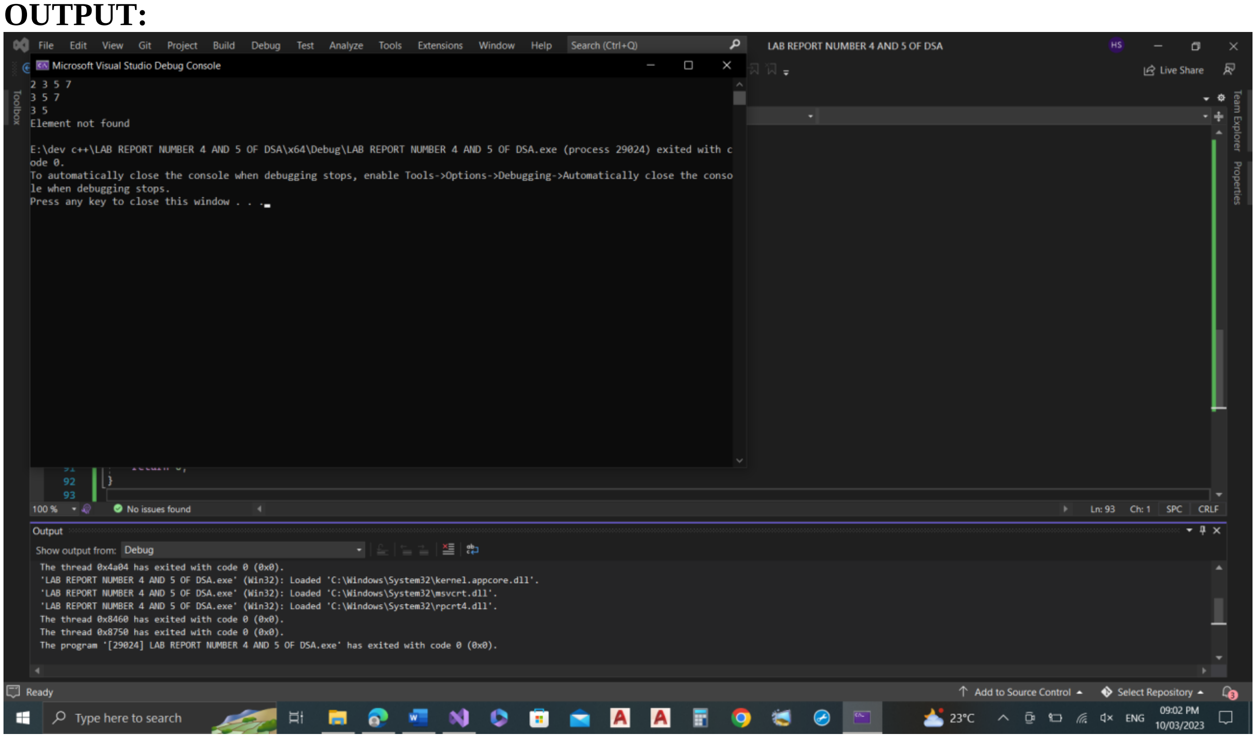
Procedure:
Singly linked list:
A singly linked list is a data structure that consists of a sequence of nodes, where each node stores a value and a pointer to the next node in the sequence. The first node in the sequence is called the head node, and the last node is called the tail node, which has a null pointer as its next node.
Sorted Singly linked list:
A sorted singly linked list in C++ is a data structure that stores a collection of elements in a linear order. Each element in the list, except for the last one, has a pointer to the next element in the list. The elements are stored in ascending or descending order based on their values.
Circular Singly linked list:
A circular singly linked list is a linked list in which the last node points to the first node instead of a null value. This creates a circular structure that can be traversed indefinitely.

Tasks

Task 1:
Write a C++ program to create Sorted Singly Linked List ADT. Your Linked List must be able to perform the following functionalities:
(a) Insertion of an integer (start/middle/end) in sorted linked list.
(b) Deletion of a given integer from the above linked list.
(c) Display the contents of the above list after deletion.
CODE:



OUTPUT:



Task 2:

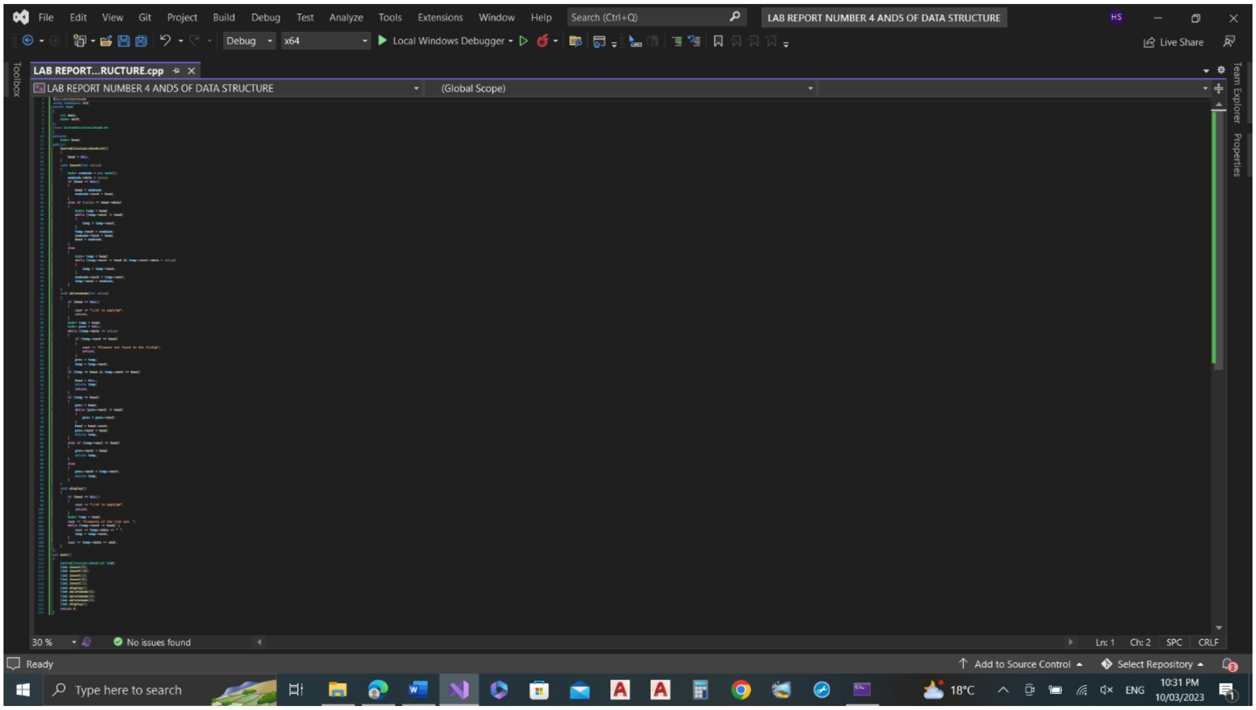
Write a C++ program to create sorted Circular Linked List ADT. Your Linked List must be able to perform the following functionalities:

(a) Insertion of an integer (start/middle/end) in sorted linked list.

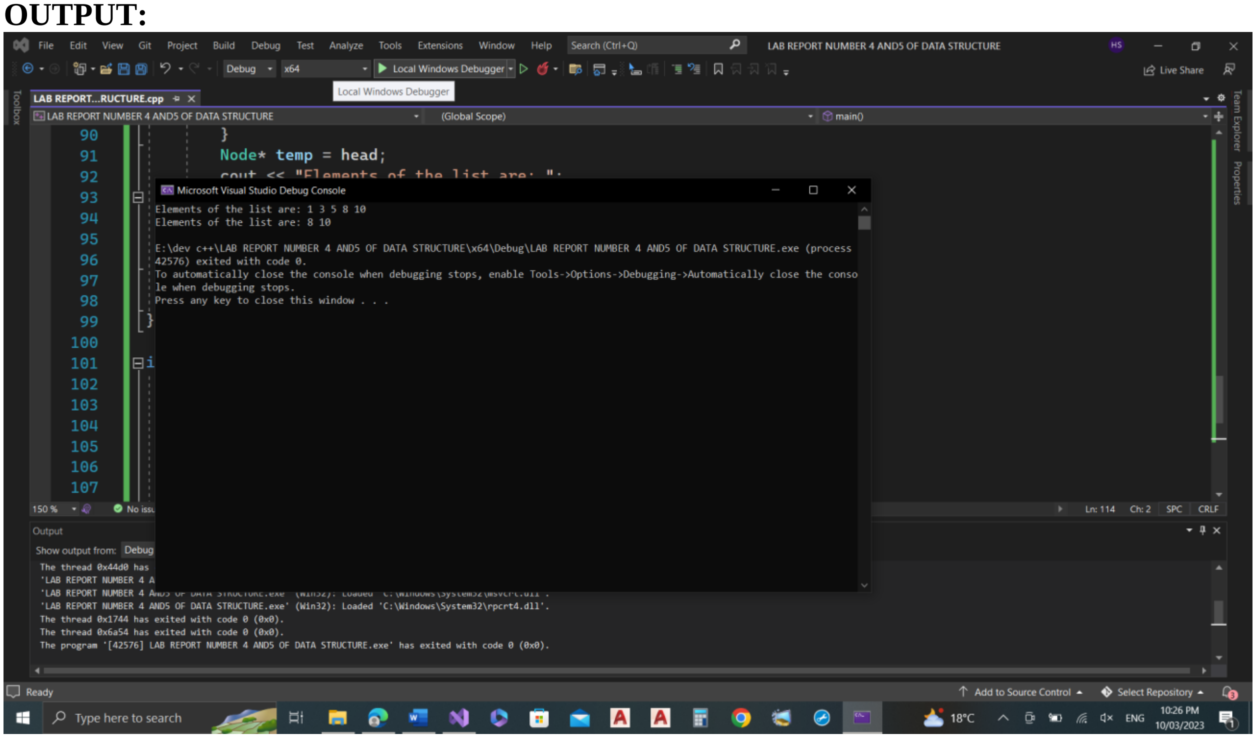
(b) Deletion of a given integer from the above linked list.

(c) Display the contents of the above list after deletion.

CODE:



OUTPUT:



Conclusion:

A **singly linked list** is a data structure composed of nodes, each containing a value and a reference to the next node in the sequence. In a **sorted singly linked list,** the nodes are arranged in ascending or descending order based on their values. In a **circular singly linked list**, the last node in the sequence points back to the first node, creating a loop.